Advancing Today's Training Defeating Tomorrow's Threats

TSMO Threat Operations

EW In-Depth

U.S. ARMY

Threat Electronic Warfare Capabilities

MANEUVER AND NETWORK
Threat Battle Command Force (TBCF)

ELECTRONIC ATTACK
Enhanced Controlled Signal Transmitter (E-CST)
Cyber and Electronic Warfare Jammer Asset (CEWJA)
Cicada-C
Wideband Configurable Controlled Jammer (WCCJ)
TRC-274
Threat High Output Range Extender (THORE)
Mission Adaptable Multi-Band Asset for GNSS (MAMBA-G)

SIGINT / DF
Advanced Networked Electronic Support Threat Sensors (AdvNESTS)
Networked Electronic Support Threat Sensors (NESTS)
Electronic Warfare Open-air Collector (EWOC)

INTEGRATED AIR DEFENSE SYSTEMS
Giraffe Agile Multi-Beam (G-AMB)
Giraffe 1X (G1X)
Giraffe 75 (G75)
Anti-Aircraft Artillery Simulator (AAA2)
Radio Frequency Surface-to-Air Missile (RF SAM)
Radar Signal Emulator (RSE)

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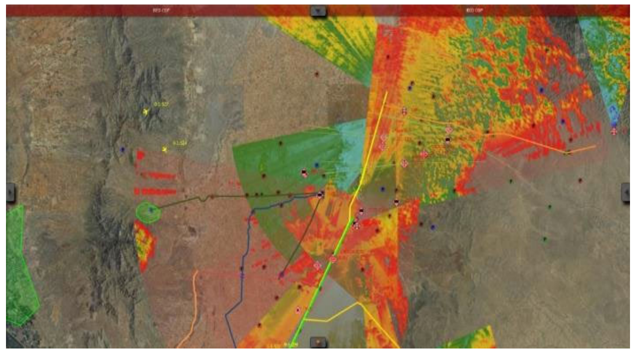
Integrated Network Command and Control

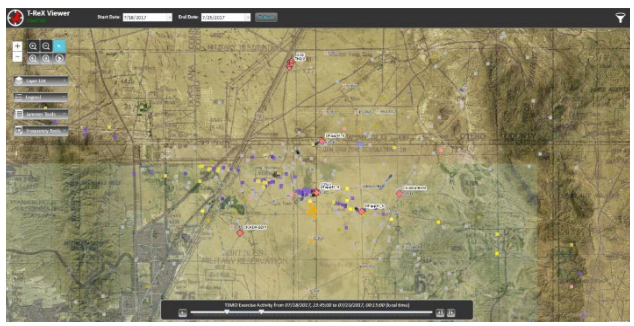
Threat Battle Command Force (TBCF)

Built on **Test and Training Enabling Architecture (TENA)**
Allows easy translation and integration with other simulation languages

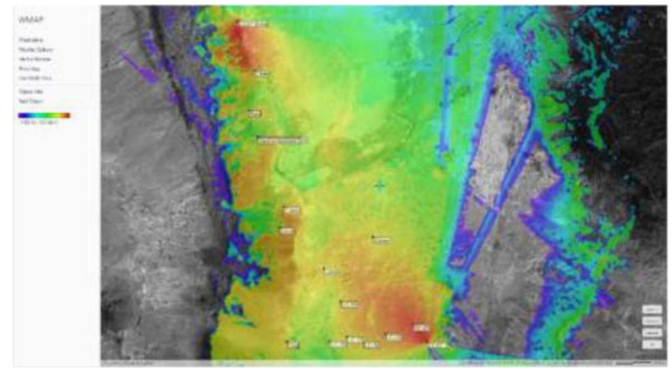
Adaptable Range Exercise System (ARES)
Real-time operational Picture

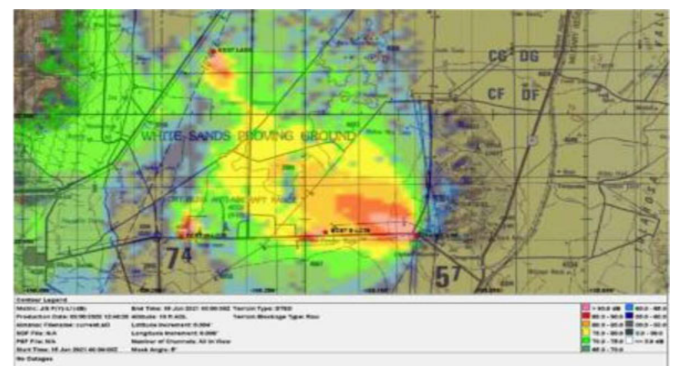


TSMO Replay Exercise Viewer (T-ReX)
Data recording and replay



RF Propagation Modelling and Analysis
Spectrum-E (general purpose) and GIANT (GPS)





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Threat Battle Command Force (TBCF)

System Description

- Threat Command, Control and Communications Suite
- Integrated Threat Situational Awareness (SA) visualization
- Command and Control (C2)
- Common operating picture (COP) at multiple locations

Integrated with:

- Electronic Warfare (EW) assets
- Cyber Red Team
- Live, Virtual and Constructive (LVC) representations
- Traditional threat maneuver and fires assets

Components:

- ARES Common Operational Picture
- TReX Historical data collection and viewer
- TCP/IP network backbone over RF, copper, or fiber
- Data collection and analysis suite
- Test Enabling Network Architecture (TENA) interconnections
- Voice via Voice over IP (VoIP) or VHF / Instant messenger chat
- Shared Data Storage

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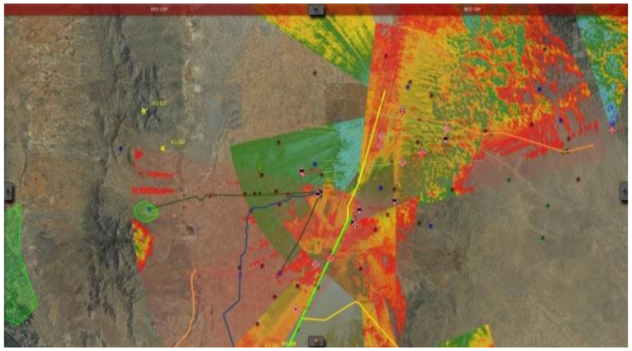
Visualization, Analysis, and Data Collection



Adaptable Range Exercise System (ARES)

- Produced by General Dynamics
- Real-time operational picture
- Visualization of:
- Platform location and movement
- NESTS DF tracks and LOB's
- RADAR tracks
- Jammer/EA activity
- Sensor System Detections

Test and Training Enabling Network Architecture

- Enables Interoperability among systems, facilities, simulations and C4ISR systems
- Adapters translate platform data and tracks into common TENA Object Model Language
- Visualization/C2 tools can be easily duplicated anywhere on the TBCF network
- New or improved Systems can be easily integrated into existing TENA framework

TSMO Replay Exercise Viewer (T-ReX)

- Based on ArcGIS, produced by ESRI
- Allows time-integrated viewing and filtering of:
- NESTS DF tracks
  - Filter by Frequency
  - Color-code by identified unit
- Jammer fans

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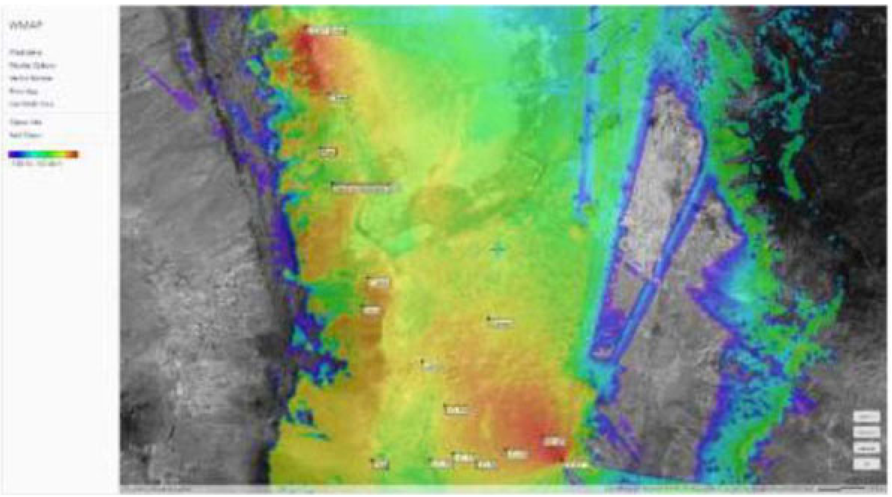
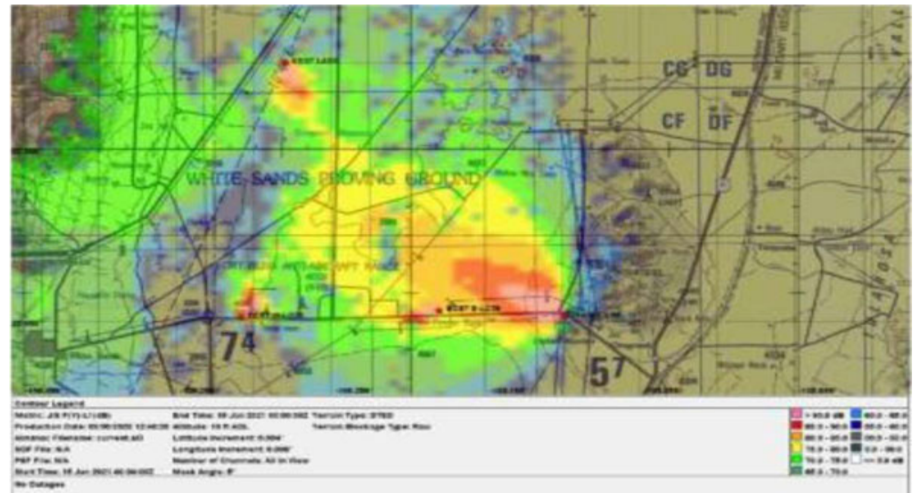
RF Propagation Modeling and Analysis





Spectrum-E / HTz Warfare

- Produced by ATDI
- Calculates:
- Jammer power-on-target
- Jammer-to-Signal Ratio
- DF Miss Distance
- Communications Signal Visibility Area / Power on Target
- Communication Link Budget

GPS Interference and Navigation Tool (GIANT)

- US DoD-owned GPS analysis tool, produced by LinQuest
- Calculates Jammer-to-Signal and Repeater-to-Signal Ratios
- Used to obtain GPS emissions effects areas and permits

Electronic Warfare Open-Air Collector (EWOC)

System Description

- RF detection and analysis laboratory
- Validates propagation modelling data for Electromagnetic Surveillance Vulnerability Assessment (ESVA) of BLUFOR units
- Provides ground truth on electronic warfare (EW) emissions
- Ensures compliance with FCC and Army guidelines for electronic attack (EA) emissions
- Confirms emitter direction (truck mounted only)

CONOPS

- Deployed near an EA system to verify emissions
- Deployed near system under test to verify EA power on target
- Used as a white-cell element to determine electromagnetic signature of target units, to assist in signature mitigation
- ***NOT A THREAT REPRESENTATIVE SYSTEM***

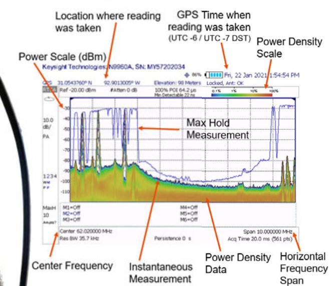
EWOC Performance

- Real-time Signal Intelligence and Analysis
- Direction Finding
- Recording and playback of intermittent and pulsed signals
- Internal amplification of low-level signals

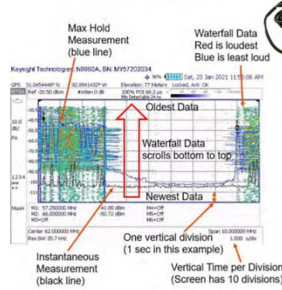
EWOC-P Performance

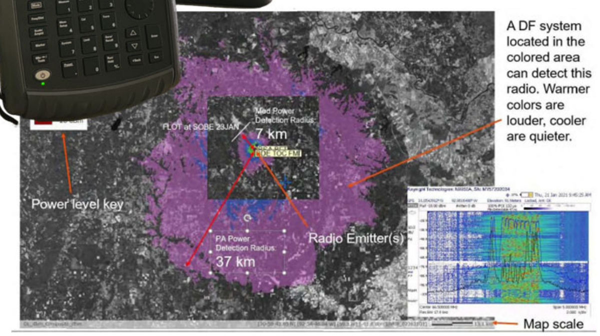
- Real-time Signal Intelligence and Analysis
- Internal amplification of low-level signals

Spectrum Density View



Waterfall View





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Advanced Networked Electronic Support Threat Sensors (AdvNESTS)

System Description

- Direction Finding (DF) using Angle and Time Difference of Arrival (AoA / TDoA)
- Frequency Correlation
- Signal Exploitation
- DF lines-of-bearing
- Emitter geo-location
- Net construction
- Signal analysis
- Provides an advanced communications detection infrastructure with Operational Coverage: 4 to 400 km² , depending on terrain

CONOPS

- Provides electronic signal and DF data for the Threat Commander and Staff.
- Information displayed at EW TOC includes frequency, emitter location, detected power, track quality
- When correlated with known unit frequencies, yields Blue Force unit locations

Performance

- Dedicated digital DF receiver systems
- Robust digital radio network with redundant data options
- High capability against modern systems and waveforms including DSSS and FHSS signals

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Networked Electronic Support Threat Sensors (NESTS)

U.S. ARMY

System Description
- Direction Finding (DF)
- Signal Exploitation
- Provides an advanced communications detection infrastructure. May be operated stand-alone or cooperatively networked.
- DF lines-of-bearing
- Emitter geo-location
- Net construction
- Signal analysis

Performance

NESTS-High
- Dedicated digital DF receiver systems
- Robust digital radio network with redundant data options
- High capability against modern systems and waveforms including DSSS and FHSS signals

NESTS-Medium
- Dedicated digital receiver only
- Digital data radio with redundant options
- Good capability against current tactical radios and cell phones

CONOPS
- Provides electronic signal and DF data for the Threat Commander and Staff.
- Information displayed at EW TOC includes frequency, emitter location, detected power, track quality
- When correlated with known unit frequencies, yields Blue Force unit locations

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Enhanced Controlled Signal Transmitter (ECST)

U.S. ARMY

System Description

- Electronic attack (EA) of Communications and Unreliability – Current Threat
- Highly Mobile
- Capable of full remote operation though ITF network
- Remote system diagnostics
- Able to control Remote Jammer Units (RJUs) for Injection Jamming

CONOPS

- Using targeting information from electronic sensors and other intelligence assets, the Threat Commander can request a local or remote operator to conduct EA on a single simultaneous frequency.
- E-CST can find additional frequencies of interest in the area, resulting in electronic attack against those targets.

Performance

- Remote Controllable through ITF network
- Setup time: 30 min - 2 hrs.

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Cyber and Electronic Warfare Jammer Asset (CEWJA)

System Description

- Software Defined Jammer
- Reprogrammable, multi-functional EA support
- Smart electronic attack
  - ESM + ECM focuses emitted power on the targets of interest to optimize the jamming effects
- Fast triggering and wide band receiving
- Effectively jams all kinds of modern communications, including frequency hopping combat net radios

CONOPS

- Using targeting information from electronic sensors and other intelligence assets, the Threat Commander can request CEWJA conduct Electronic Attack (EA) on multiple frequencies in different bands
- CEWJA can find additional frequencies of interest in the area resulting in EA against those targets
- CEWJA can automatically jam frequencies upon detection

Performance

- Smart ESM and ECM capabilities
- Multirange: HF, V/UHF, SHF, and C-Band
- Library of jamming waveforms
- Custom waveform creation software
- Follower-jammer
- DRFM repeater

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Cicada-C

System Description

- Jams up to 16 active radio nets simultaneously
- Example targets: Frequency hoppers, cell phones, satellite navigation
- Reactive jam modes scan the environment and activate only when target signals are present

CONOPS

- Using targeting information from electronic sensors and other intelligence assets, the Threat Commander can request CICADA conduct Electronic Attack (EA) on multiple frequencies in different bands
- CICADA can find additional frequencies of interest in the area resulting in EA against those targets
- CICADA can automatically jam frequencies upon detection

Performance

- Automatic, computer controlled jamming sequences
- High power, liquid cooled power amplifiers
- Extremely fast multiplex operation for effective jamming of multiple targets simultaneously
- Special jamming modulations ensuring effectiveness
- Jamming of predefined frequencies in order of priority
- Barrage jamming against several frequency hoppers
- Protected frequencies to be spared from jamming
- Low false alarm rate and high probability of intercept
- Setup time: 1-2 hours

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Wideband Configurable Controlled Jammer (WCCJ)

System Description

- EA of Communications and Unreliability – Current Threat
- Example Targets: *FM Voice, SRW / WNW, Satellite Communications*
- Minimal out-of-band emissions, even near restricted frequencies
- Replicates many different Threat jamming systems

CONOPS

- Using targeting information from electronic sensors and other intelligence assets, the Threat Commander can request the WCCJ to conduct electronic attacks (EA) on 2 different frequencies simultaneously.
- WCCJ can find additional frequencies of interest in the area resulting in EA against those targets.

Performance

- Frequency follower plus other Direct Sequence Spread Spectrum (DSSS) threat jamming capabilities
- Digital Radio Frequency Memory (DRFM) capability
- Direction Finding (DF) capability
- Wideband frequency output
- 2 independent output channels allow simultaneous jamming of 2 separate frequencies
- Dynamically Programmable Output Waveforms
- Flexible output frequency tailoring
- Remote monitor capability using the ITF network
- Setup time: 1-2 hours

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TRC-274 Software Defined Radio Jammer

System Description

- Software Defined Jammer
- Reprogrammable, multi-functional EA support
- Smart electronic attack
  - ESM + ECM focuses emitted power on the targets of interest to optimize the jamming effects
- Fast triggering and wide band receiving
- Effectively jams all kinds of modern communications, including frequency hopping combat net radios

CONOPS

- Using targeting information from electronic sensors and other intelligence assets, the Threat Commander can request TRC-274 conduct Electronic Attack (EA) on multiple frequencies in different bands
- TRC-274 can find additional frequencies of interest in the area resulting in EA against those targets
- TRC-274 can automatically jam frequencies upon detection

Performance

- Smart: ESM and ECM capabilities
- Fast trigger time
- Multirange: HF, V/UHF, and SHF
- High performance
- Library of jamming waveforms
- Spot and Responsive jamming
- Deception
- Smart chirp
- Follower-jammer
- DRFM repeater
- Setup time: 1-2 hours

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Threat High Output Range Extender (THORE)

System Description

- Unmanned Network Controlled PNT Jammer System
- Electronic attack (EA) of PNT signals using current threat techniques
- Capable of full remote operation though TBCF network
- Remote system diagnostics

CONOPS

- Sophisticated, programmable radio frequency (RF) threat emulator
- Provides high fidelity simulation of known Position, Navigation and Timing (PNT) denial threats
- Designed to assess operational capabilities in Global Navigation Satellite Systems (GNSS) degraded/denied environments

Performance

- Remote Controllable through TBCF Network
- Setup time: 30 min – 1 hr.

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Mission Adaptable Multi-Band Asset for GNSS (MAMBA-G)

System Description
- Unmanned Network Controlled PNT Jammer System
- Electronic attack (EA) of PNT signals using current, emerging, and currently-emerging threat techniques
- Capable of full remote operation though TBCF network
- Remote system diagnostics

CONOPS
- Sophisticated, programmable radio frequency (RF) threat emulator
- Provides high fidelity simulation of known Position, Navigation and Timing (PNT) denial threats
- Designed to assess operational capabilities in Global Navigation Satellite Systems (GNSS) degraded/denied environments

Performance
- Remote Controllable through TBCF Network
- Setup time: 30 min – 1 hr.

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Giraffe Agile Multi-Beam (GAMB)

U.S. ARMY

System Description

- Multi-beam 3-Dimensional air coverage
- Frequency agile pulse Doppler air search radar
- Mobile or static short to medium range air defense application
- Detects hovering helicopters
- Detects in-coming artillery and mortar fire
- Detects low-altitude, low cross-section aircraft targets in conditions of severe clutter and electronic countermeasures

CONOPS

- Air surveillance and tracking
- Fixed-wing and Rotary-wing tracking
- Rocket, Artillery, and Mortar Tracking
- Surface surveillance and tracking
- Target identification for weapon systems
- All tracks displayed real-time on COP via ITF network

Performance

- Multi-beam 3-Dimensional air coverage with instrumented ranges
- Operational Frequency range
- Data rate 1-scan-per-second
- Pulse density suppresses high cluttering in adverse weather.
- Ultra-low antenna side-lobes combined with pulse-to-pulse and burst-to-burst frequency agility provides some resistance to jamming
- Automatic Point of Origin / Point of Impact calculations
- Deployable in 10 minutes, prepared to travel in 6 minutes.

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Giraffe 1X (G1X)

System Description
- Newest member of the Giraffe family of radar
- Automated Threat Evaluation and Weapons Assignment
- Target data and fire control orders for allocated weapons
- Distribution of target data over tactical data links
- Detects hovering helicopters
- Detects in-coming artillery and mortar fire
- Detects low-altitude, low cross-section aircraft targets in conditions of severe clutter and electronic countermeasures

CONOPS
- Air surveillance and tracking
- Fixed-wing and Rotary-wing tracking
- Rocket, Artillery, and Mortar Tracking
- Surface surveillance and tracking
- Target identification for weapon systems
- All tracks displayed real-time on COP via ITF network

Performance
- Multi-beam 3-Dimensional air coverage
- X Band, Multi Channel
- Simultaneous tracking of surface and air tracks
- Data rate 1-scan-per-second
- Pulse density suppresses high cluttering in adverse weather.
- Ultra-low antenna side-lobes and frequency hopping with auto selection of least jammed frequency

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Giraffe 75

System Description

- Frequency agile pulse Doppler air search radar
- Mobile or static short to medium range air defense application
- Detects low-altitude, low cross-section aircraft targets in conditions of severe clutter and electronic countermeasures

CONOPS

- Air surveillance and tracking
- Surface surveillance and tracking
- Target identification for weapon systems
- Can provide an air picture to each firing battery using FM radio communication

Performance

- 13-meter antenna mast
- Includes an integrated Command, Control and Communication (C3) system that enables GIRAFFE to act as the command-and-control center in an air defense system

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Anti-Aircraft Artillery Simulator (AAA2)

U.S. ARMY

System Description

- Ground-based air defense threat emitters
- Create a realistic electronic warfare environment for training fixed- and rotary-wing aviators
- Daylight and FLIR cameras for target acquisition, RADAR pulse emitter to stimulate RWS

Emulated 23-mm SP Anti-aircraft Artillery

Day and Night Vision

On-Board Radar

- Detection
- Tracking

23 mm AA guns

Emulated 30-mm SP Anti-aircraft Artillery

Day and Night Vision

Dual Tracking / Detection Radars

2 x 30 mm twin barrel cannons

- ***Surface-to-air missiles***
- Must stop to fire missiles

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Radio Frequency Surface-to-air Missile Simulator (RF-SAM)

U.S. ARMY

System Description

- Ground-based air defense threat emitters
- Create a realistic electronic warfare environment for training fixed- and rotary-wing aviators
- Daylight and FLIR cameras for target acquisition, RADAR pulse emitter to stimulate RWS

Armament

Surface-to-air missile

Radars

Target Acquisition Radar

Mono-pulse Target Tracking Radar

Missile Tracking

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CAPABILITY PROGRAM EXECUTIVE
CPE ST3
SIMULATION, TRAINING, TEST & THREAT

Thank you